Finite carrier bounds for signed arithmetic energy

A proof of the bare source logarithmic sixth power bound and a conditional repeated decorator refinement

Abstract

The normalized arithmetic source formed from two von Mangoldt carriers and two truncated Möbius long roles has cumulative squared energy $O(T \log^6 T)$. The essential finite-window estimate cancels the carrier divisor mass against the original logarithmic normalization before summing. Combined with the joint-row harmonic fifth-moment bound and the lower-order large-prime sectors, it controls the complete bare arithmetic source, including ordered carrier crosses and repeated prime powers.

A separate first-power core sector admits a logarithmic fifth-power bound under explicitly stated uniform marked-role and physical-map inputs. Its repeated decorating prime is selected above a scale-dependent threshold $A(x)$, not an absolute constant. The internal cross between its two packets has exponent 9/2; an overlapping external sixth-power field has only the generic exponent 11/2 cross.

The paper separates analytic proofs, inherited hypotheses, exact finite checks, and unresolved transfers. It makes no claim about an unspecified terminal response, complete fifth-power energy, a sharp physical lower bound, or a leading constant. The finite audit packet checks source algebra and boundary sensitivity, rather than certifying an asymptotic theorem.

1 Scope and conventions

All logarithms are natural. Dirichlet convolution is $(a*b)(n)=\Sigma_{de=n}a(d)b(e)$. The Möbius function is μ, the von Mangoldt function is Λ, and δ_1 is the unit for convolution. An integer is U-smooth when all of its prime factors are at most U. All constants in O notation are fixed before the scale varies.

The principal statement concerns a fully defined arithmetic observable. Its derivation uses a fixed-precision modulus-one Möbius estimate from an established external theorem, together with elementary positive estimates. Sections 4–5 state the imported input and the harmonic proof. Section 7 records a different, conditional sector theorem without identifying its source with the full observable.

This is a retrospective reconstruction and audit of dated research arguments [1–7]. It is neither a preregistered experiment nor a literature novelty assessment. Historical exponent-seven and exponent-eight labels in predecessor notes describe earlier bounds, not the present frontier.

2 The observable and its exact window

For real $T>4096$ put $U=2T^{1/3}$, $L_0=\log U$, and define

$$C(n)=\Sigma_{d|n,\ d\le U}\ \mu(d)-\delta_1(n),\qquad C_{\mathrm{long}}(n)=C(n)\,1_{n>U},$$

$$\Lambda_{\mathrm{short}}(n)=\Lambda(n)\,1_{n\le T^{\wedge}(1/3)},$$

$$F_T(N)=3(\Lambda^*\Lambda_{\mathrm{short}}{}^*C_{\mathrm{long}}{}^*C_{\mathrm{long}})(N)/\log N,\qquad 1<N\le T. \qquad (1)$$

For $T\le 4096$ the source is zero. Indeed its nonzero factors would require two carriers of product at least 4 and two long roles of product greater than U^2; thus $N>4U^2=16T^{2/3}$, which is incompatible with $N\le T$ in that range.

Let $j_U(N)=\Sigma_{p>U}v_p(N)$. Since $U^3=8T$, every $N\le T$ has $j_U(N)\in\{0,1,2\}$. These are disjoint complete output sets, so their energies E_0,E_1,E_2 add exactly. The word “output” is essential: internal convolution roles are not orthogonal simply because their descriptions differ.

Let f_U be C_{long} restricted to U-smooth integers, let $D_U=f_U{}^*f_U$, and let $K=\Lambda^*\Lambda$. On $j_U(N)=0$ the source is

$$F_0(N)=3\ \Sigma_{ms=N}K(m)D_U(s)/\log N. \qquad (2)$$

Every actual term satisfies

$$U^2<s\le T/4,\qquad 4\le m<T/U^2=U/8. \qquad (3)$$

Each carrier factor is at most $m<U/8<U/2=T^{1/3}$. Thus the individual short-carrier cutoff in (1) is automatic on this sector; no cutoff has been replaced.

For any coefficient sequence D supported on U-smooth $s>U^2$, write

$$N_h(D)=\Sigma_s\ |D(s)|^2/s.$$

The upcoming carrier lemma applies whenever this norm is finite, and otherwise gives a vacuous extended-real upper bound. It does not cover a history-dependent mask that changes with the carrier unless that mask is included in a newly defined and controlled response.

Repeated carriers remain literal. For example, $K(2^j)=(j-1)(\log 2)^2$ for $j\ge 2$. In the exact Gram parametrization $m_1=da$, $m_2=db$, $s_1=bk$, $s_2=ak$, only $\gcd(a,b)=1$ is imposed. The common divisor d may share primes with a,b,k. In particular, the pair 4 and 8 has $d=4,a=1,b=2$ and is admissible.

3 The finite carrier lemma

Theorem 1. Suppose $\psi(z)=\Sigma_{n\le z}\Lambda(n)\le C\psi z$ for all $z\ge1$, with a fixed constant $C\psi$. For the response in (2), with any D satisfying the support in Section 2,

$$\Sigma_{N\le T}|F_D(N)|^2 \le 9\, C\psi^2\, T\, \Sigma_{U^2<s\le T/4}\, |D(s)|^2[1+\log(T/s)]/s$$
$$\le 9\, C\psi^2\, T(1+L_0)N_h(D). \qquad (4)$$

All carrier crosses are included in this inequality.

Proof. Define the admissible divisor mass $A_U(N)=\Sigma_{m|N,\ N/m>U^2}K(m)$. Positivity and the identity $\Sigma_{a|N}\Lambda(a)=\log N$ give

$$A_U(N) \le \Sigma_{ab|N}\Lambda(a)\Lambda(b) \le (\Sigma_{a|N}\Lambda(a))^2 = (\log N)^2.$$

Weighted Cauchy applied to the complete signed sum, before summing over outputs, yields

$$|F_D(N)|^2 \le 9\, A_U(N)/(\log N)^2 \cdot \Sigma_{ms=N}K(m)|D(s)|^2$$
$$\le 9\, \Sigma_{ms=N}K(m)|D(s)|^2. \qquad (5)$$

The normalization cancels the divisor mass here. Bounding the normalizer separately first would lose this gain.

Positive summation and partial summation give

$$\Psi_2(z)=\Sigma_{m\le z}K(m) \le C\psi z\, \Sigma_{a\le z}\Lambda(a)/a$$
$$\le C\psi^2\, z(1+\log z).$$

Summing (5), and retaining the nonempty carrier window $s\le T/4$, proves the first line of (4). Since $T/s<U/8<U$, the second line follows by positive enlargement. This completes the proof.

The exact signed Gram before applying Cauchy is

$$9\, \Sigma\, K(da)K(db)D(bk)\mathrm{conj}(D(ak))/(\log(dabk))^2, \qquad (6)$$

where $d,a,b,k\ge1$, $\gcd(a,b)=1$, $dabk\le T$, and both cofactor and smooth-output supports are retained. Equation (6) has both cross orientations. The proof does not replace its signs by an assumption of cancellation or independence.

Corollary 1. If $N_h(D_U)\le C_f(1+L_0)^5$, then $E_0\le 9C\psi^2C_f\, T(1+L_0)^6$. The next two sections derive the harmonic hypothesis at fixed imported precision. The further large-prime sector estimates in Section 6 complete the source theorem.

4 The harmonic input and scalar cancellation

On the normalized torus over primes $p \le U$ put $x_p = z_p/\sqrt{p}$ and

$$Z_U = \prod_{p \le U} (1 - x_p)^{-1}, \quad B_U = \Sigma_{d \le U} \mu(d) x_d, \quad f_U = Z_U\, B_U - 1.$$

Here $x_d = \prod_p x_p^{v_p(d)}$. The arithmetic coefficient of f_U at n is the U-smooth restriction of C(n). Its unit is zero, and all coefficients for $1 < n \le U$ vanish by $\mu * 1 = \delta_1$. Thus it is the same long role as in Section 2. Parseval gives $N_h(f_U * f_U) = \|f_U\|_4^4$.

The fixed external input is $|M(X)| \le C_4 X/(1+\log X)^4$ for every $X \ge 1$, with $M(X) = \Sigma_{n \le X} \mu(n)$. It follows by fixing A=4 in Proposition 3 of Murty and Vatwani [1], then retaining only modulus one and absorbing the finite initial range. Precision and constants do not grow with U or any excluded prime set.

Partial summation gives a uniformly convergent tail for $\Sigma\, \mu(n)/n^\sigma$ as $\sigma \downarrow 1$. For $1 \le \sigma \le 2$ its tail after y has absolute value at most

$$C_4[(1+\log y)^{-4} + (2/3)(1+\log y)^{-3}].$$

For $\sigma > 1$ this series is $1/\zeta(\sigma)$, which tends to zero. Hence $\Sigma\, \mu(n)/n = 0$ and, for a fixed C,

$$|M_1(y)| = |\Sigma_{n \le y} \mu(n)/n| \le C/(1+\log y)^2. \qquad (7)$$

This proves, rather than assumes, the needed harmonic cancellation.

Write $F_1(x) = \Sigma_{ab \le x} \mu(a)\mu(b)/(ab)$. The inclusive hyperbola identity

$$F_1(x) = 2\Sigma_{a \le \sqrt{x}}\, \mu(a) M_1(x/a)/a - M_1(\sqrt{x})^2$$

and (7) imply $|F_1(x)| \le K_F/(1+\log x)$ for a fixed K_F. Let $w(p) = 2/(p+1)$, and $S_r(x) = \Sigma_{a \le x,\ (a,r)=1} \mu(a) w(a)$. The coefficient identity

$$S_1(x) = \Sigma_{d \le x} e(d) F_1(x/d)/d,$$

$$e(p^j) = 1 - j(p-1)/(p+1)$$

has $\Sigma |e(d)| d^{-3/4} < \infty$. Splitting d at $\sqrt{x}$ gives $|S_1(x)| \le K_S/(1+\log x)$. This is a weighted Möbius estimate, not a uniform bound on arbitrary signed rows. The remaining exclusion masses are kept explicitly in Section 5.

5 The joint row fifth moment

Under $d\nu=|Z_U|^4 d\theta/M_U$, where $M_U=\|Z_U\|_4^4\le C_0^4(1+L_0)^4$, the orthogonal local basis is $e_0=1$, $e_1=x-w$, $e_2=(x-t)^2$, with $t=1/p$. Its squared norms and conversion formula are

$$d_1=t(1-t)^2/(1+t)^2, \quad d_2=t^2(1-t)^3/(1+t),$$

$$x^2=e_2+2te_1+m_2, \quad m_2=t^2(3-t)/(1+t).$$

The bound for M_U follows from the classical elementary Mertens product bound; C_0 is fixed. Finite prime products suffice throughout, and the correction products below converge uniformly.

For disjoint squarefree g,h let $C_{g,h}$ be the coefficient with e_1 on g and e_2 on h in B_U^2. Orthogonality gives

$$\|Z_U\, B_U\|_4^4=M_U \sum_{(g,h)=1} d_1(g)d_2(h)|C_{g,h}|^2. \tag{8}$$

Set $\kappa(p)=-(1-1/p)^2/4$, $\delta=1/4$, $H_r=\prod_{p\mid r}(1-w(p))^{-1}$ and $H_{r,\delta}=\prod_{p\mid r}(1-w(p)p^{\delta})^{-1}$. The signed rectangular kernel is

$$K_r(X,Y)=\sum_{d\le \min(X,Y),(d,r)=1}\kappa(d)w(d)^2\, S_{rd}(X/d)S_{rd}(Y/d).$$

It is zero if $X<1$ or $Y<1$. Positive exclusion convolution, full-mass Jensen, and the bound for S_1 imply

$$|K_r(X,Y)|\le K\theta\, H_r\, H_{r,\delta}/[(1+\log X)(1+\log Y)]. \tag{9}$$

For completeness, the positive averaged kernel after exclusion has prime-power coefficients $\beta(p^j)=(p-1)w(p)^{j-1}/(p^2(p+1))$. Its δ moment $K\beta$ is finite. A $\sqrt{x}$ split bounds $\sum_{n\le x}\beta(n)/(1+\log(x/n))^2$ by $260K\beta/(1+\log x)^2$. Taking $K\theta=260K_S^2K\beta$ proves (9) by Cauchy; no H_r factor is dropped.

The exact allocation of first-order primes among the two Möbius divisors is

$$C_{g,h}=\sum_{abc=g}\mu(a)\mu(b)\zeta(c)K_{gh}(U/(hac),U/(hbc)), \tag{10}$$

where a,b,c are pairwise coprime and $\zeta(p)=2/p$. State-two primes belong to both divisors. In particular, there is no extraneous $w(h)^2$ factor. Both original rectangular cutoffs remain coupled in (10).

5 Continued The positive endpoint bound

Define $f_S(n)=1_{n\le S}/(1+\log(S/n))$, $R_S=f_S*f_S$, and let $V(n)=\prod_{p\mid n}v_p$ with $v_p\ge1$ and $A_V=\prod_p(1+(v_p-1)/p)<\infty$. Then

$$\Sigma_k\, V(k)R_S(k)^2/k \le 1536\, A_V^4(1+\log S). \qquad (11)$$

Proof. The unique parametrization of $ab=cd$ is $a=\alpha r, b=\beta s, c=\alpha s, d=\beta r$ with $\gcd(r,s)=1$. Bound $V(\alpha\beta rs)$ by $V(\alpha)V(\beta)V(r)V(s)$, then drop only positive coprimality restrictions. With $m=\max(r,s)$, $n=\min(r,s)$, $X=S/m$ and $\Delta=\log(m/n)$, each α or β sum is bounded by $8A_V\ell(\Delta)$, where $\ell(t)=(1+\log(1+t))/(1+t)$.

To obtain this bound first partition j by $\log(X/j)$ in half-open intervals of length one. Each interval has harmonic mass at most 2. Integral comparison bounds the resulting sum of $1/[(1+k)(1+k+\Delta)]$ by $4\ell(\Delta)$. Writing $V=1*e$ with $e(p)=v_p-1\ge0$ and $\Sigma e(d)/d=A_V$ transfers the estimate to V. Also, $\int_0^\infty \ell(t)^2dt=5$, so $\Sigma_{n\le m}V(n)\ell(\log(m/n))^2/n\le12A_V$. The remaining sum over m is $\le A_V(1+\log S)$. The two possible orders of r,s give $2\cdot8^2\cdot12=1536$, proving (11).

Apply weighted Cauchy to the c-allocation in (10), retaining its full mass $P_g=\prod_{p\mid g}(1+2/p)$. The exact local identity $d_1(p)H_p^2=1/p$ leaves the positive bound

$$\Sigma_{g,h}d_1(g)d_2(h)|C_{g,h}|^2$$

$$\le K\theta^2\,\Sigma_h\,\gamma(h)\,\Sigma_{c\ \mathrm{squarefree}}\,q(c)\zeta(c)\,E_q(U/(hc))/c, \quad (12)$$

where E_q is the left side of (11), $q_p=H_{p,\delta}^2(1+2/p)$, and $\gamma(h)=d_2(h)H_h^2H_{h,\delta}^2$. Empty arguments contribute zero. Restrictions are enlarged only after positivity.

Since $q_p-1=O(p^{-3/4})$, both A_q and $C_c=\prod_p(1+2q_p/p^2)$ converge. Also $d_2(p)H_p^2=p^{-2}(1-p^{-2})$, so $K\gamma=\Sigma_h\gamma(h)<\infty$. Equations (11)–(12) give the bound $K_R(1+L_0)$, with $K_R=1536K\theta^2A_q^4C_cK\gamma$. Finally the unit is retained by norm triangle:

$$\|f_U\|_4^4\le8(\|Z_U\,B_U\|_4^4+1)\le C_f(1+L_0)^5,$$

$$C_f=8(C_0^4K_R+1). \qquad (13)$$

This proves the harmonic input. The argument controls the full row family, including all repeated second-order modes, rather than adding overlapping sector estimates.

6 Completion by the large prime sectors

For $j_U=1$ write $N=rn$ with $r>U$ prime and n U-smooth. A long role containing r cannot carry a nontrivial small cofactor: the other long role is $>U$, so that cofactor is $<U/8$, and its complete Möbius divisor sum vanishes unless it is one. Consequently

$$F_T(rn)=6B(n)/\log(rn),$$

$$B(n)=\Sigma_{m|n,m<U/8}K(m)C_{\mathrm{long}}(n/m).$$

The smooth harmonic role norm $S_0=\Sigma_{n\ U\text{-smooth}}|C_{\mathrm{long}}(n)|^2/n$ has the exact representation $S_0=P_U\ Q_1-1$, where $P_U=\prod_{p\le U}(1-1/p)^{-1}$ and

$$Q_1=\Sigma_{g\le U,\ \mathrm{squarefree}}w_1(g)M_g(U/g)^2,$$

$$w_1(p)=p^{-1}(1-p^{-1}),\qquad M_g(t)=\Sigma_{a\le t,(a,g)=1}\mu(a)/a.$$

Positive exclusion convolution and full-mass Cauchy imply $Q_1\le\Sigma_{n\le U}M_1(U/n)^2/n$. The local coefficient collapses to p^{-k} at $n=p^k$; no modulus-independent coprime estimate is assumed. Equation (7) and ratio-e intervals give $Q_1\le(8/3)C^2$, so $S_0=O(1+L_0)$.

On the actual $n<T/U$ window, $\Sigma_{m|n}K(m)<4L_0^2$. With the fixed elementary carrier estimate $\Sigma_{m<U/8}K(m)/m\le c_0^2L_0^2$, Cauchy gives $\Sigma|B(n)|^2/n\le 4c_0^2L_0^4S_0$. Counting r by $\pi(z)\le C\pi z/\log z$, and using $\log(T/n)>L_0$ and $\log(rn)>2L_0$, yields

$$E_1\le 36C\pi c_0^2TL_0S_0=O(T(1+L_0)^2). \qquad (14)$$

For $j_U=2$ write $N=mrs$ with $U<r\le s$ prime and $m<U/8$. Each long role must contain one large prime, and the same exact Möbius identity forces it to contain no nontrivial factor of m. Therefore

$$F_T(mrs)=3K(m)(2-1_{r=s})/\log(mrs).$$

All carrier powers are retained: $K(p^a)=(a-1)(\log p)^2$ and $K(p^a q^b)=2\log p\log q$ for distinct primes. Summing the geometric exponent series gives $\Sigma_{m<U/8}K(m)^2/m=O((1+L_0)^4)$. Ordered prime-pair counting gives $O(z/L_0)$ pairs above U with product $\le z\le T$. Together with the normalizer this proves $E_2=O(T(1+L_0))$. See [5] for explicit conservative constants.

Theorem 2. Equations (4), (13), and (14), with the two-large-prime estimate, prove $\Sigma_{1<N\le T}|F_T(N)|^2=O(T\log^6T)$. The disjointness used here is exactly that of j_U output sectors. No assertion is made about preservation of that disjointness by a later operator.

7 A separate repeated decorator sector

The following refinement concerns a specified odd y-smooth paired source, rather than a silent change in the observable of Theorem 2. Put $T=x^3/8$, $M=T/4$, $\ell=\log x$, $L=1+\ell$ and $y=\sqrt{x}$. Over odd primes at most y let $b_a(d)=\mu(d)1_{d\le a}$, $m=b_x-b_{x/2}$, $f=Zb_x-\delta_1$, $G=Zm$, $Q_U=f*G$ and $Q_B=G*G$. Here Z is the all-ones series on this smooth support.

The following hypothesis specifies exactly what is inherited. Fix a coefficient mask applied after the joint convolution Q_j; denote its result H_j. The historical choice is $1_{\kappa>K_o}Q_j$, with κ the predecessor statistic and K_o fixed. No definition of an unspecified κ is needed for the conditional theorem: its admissible masks are precisely those for which all marked-profile and physical-map bounds below hold. The unmasked case $H_j=Q_j$ is included under the same inputs.

$$T_j(n)=c_j(K*H_j)(n)/\log(4n),\quad c_U=6,\ c_B=3,$$

$$K=\Lambda_o*\Lambda_o,\quad \mathbf{T}=(\gamma T_U,T_U+T_B),\quad 1\le\gamma\le 2.$$

The norm $\mathfrak{E}_E(\mathbf{T})$ is the sum over $n\in E$ of the squared Euclidean norm of this two-component vector. Λ_o is Λ restricted to odd integers. Set the unit-output field to zero.

The class E consists of $n=pw\le M$ with p prime to exponent exactly one, $p>2P^+(w)$, $p\le y$, and at least one repeated decorating prime $q\ge A(x)$. Set $P^+(1)=1$. Select the least such q and write $e=v_q(n)\ge 2$. Other decorating prime powers are unrestricted. Define

$$C^*=\max(1,C_\Pi),\quad A(x)=(\ell-\log 2)/(4C^*),\quad s=\log A,$$

$$B=896C_\Pi,\quad u=\log L,$$

$$u\ge\max(16,(2B)^2,2\log(16C^*)+2). \tag{15}$$

Together with the inherited all-cutoff range, this implies $s\ge u/2$ and $1/A\le 16C^*/L$. The formula is fixed; its value grows with x. The range is asymptotic and potentially enormous, not a practical threshold guarantee.

Let $W(a)=\tau(a)^2 16^{\omega(a)}$, $C_{src}=16(C_a+1)^4$, and let $H_{j,b}$ denote the coefficient profile obtained by fixing the full marked-prime part b and retaining all unmarked coefficients. The uniform input is $N_h(H_{j,b})\le C_{src}\,W(p)W(q^e)L^5$ for every $b\mid pq^e$, with the analogous one-mark bound. Crucially, the complete joint coefficient is formed before masking. The physical-map hypotheses used in the next section are stated there in full; this paper does not claim to have derived those inherited inputs from scratch.

8 Conditional sector bound and its crosses

For $n=pq^e$ z let D′ be the decorating primes other than q. Splitting the two carriers according to q-divisibility gives the exact no-q-carrier packet

$$Z_j(n)=c_j[(K_D*H_{j,pq^e})(z)$$
$$+2\log p(\Lambda_{D'}*H_{j,q^e})(z)]/\log(4n).$$

For the remaining packet condition only on q, writing $n=q^e$ t. The free prime set D_q includes p. Profiles depend on the marked set: Z conditions on {p,q}, whereas J conditions only on {q}; identical profile labels in these two formulas do not identify their ambient functions:

$$J_1,j(n)=2c_j \log q \sum_{r=1}^{e}(\Lambda_{D_q}*H_{j,q^{e-r}})(t)/\log(4n),$$

$$J_2,j(n)=c_j(\log q)^2 \sum_{r,s\ge1,r+s\le e}H_{j,q^{e-r-s}}(t)/\log(4n).$$

Then $\mathbf{T}=\mathbf{Z}+\mathbf{J}_1+\mathbf{J}_2$ exactly on E. Residual q powers, the p–q carrier path, and the distinct masks with and without p all remain.

The physical-map hypotheses are the following complete conditional squares, with all relevant output restrictions retained until after taking the square:

$$\mathfrak{E}_{p,q^e}(\mathbf{Z})\le C_{01}\, M\, W(p)W(q^e)L^6/(pq^e),$$

$$\mathfrak{E}_{q^e}(\mathbf{J}_1)\le C_{mix}\, M\, W(q^e)L^4(\log q^e)^2/q^e,$$

$$\mathfrak{E}_{q^e}(\mathbf{J}_2)\le C_{two}\, M\, W(q^e)L^3(\log q^e)^4/q^e. \qquad (16)$$

Here $C_{01}=270C_{src}[\sqrt{3}\,C\psi+2\sqrt{(3C\psi)}]^2$, $C_{mix}=3240C\psi C_{src}$, and $C_{two}=270C_{src}$. The inherited support-one core assumption is $\Sigma_{\text{eligible }p}W(p)/p\le 3Bu$; that logarithmic factor must not be replaced by a smaller multi-core bound.

For $q\ge A$ and $e\ge2$ the repeat tail is $\Sigma W(q^e)/q^e\le 1248C\pi/(As)$. Its local estimate is $16\Sigma_{e\ge2}(e+1)^2/q^e\le 312/q^2$; prime partial summation supplies the remaining factor. Thus

$$(3Bu)\cdot 1248C\pi/(As)\le 119808BC\pi C^*/L.$$

The marked-power moments satisfy $\Sigma\, W(a)a^{-3/4}\le\exp(1152)$ over odd powerful a. Consequently their second and fourth logarithmic moments with weight 1/a are at most 64exp(1152) and 65536exp(1152). These are fixed Euler-product bounds, not finite experimental observations.

Theorem 3, conditional on (16), the marked-profile input and core-mass bound. Put $C_Z=119808BC\pi C^*C_{01}$ and $C_J=2\exp(1152)(64C_{mix}+65536C_{two})$. Then

$$\mathfrak{E}_E(\mathbf{Z})\le C_Z\, ML^5, \quad \mathfrak{E}_E(\mathbf{J}_1+\mathbf{J}_2)\le C_J\, ML^4,$$

$$\mathfrak{E}_E(\mathbf{T})\le 2(C_Z+C_J)ML^5. \qquad (17)$$

The proof is positive owner summation followed by norm triangle. Its internal doubled Z/J cross is $\le 2\sqrt{(C_Z\, C_J)}ML^{9/2}$. A separate overlapping field with energy $O(ML^6)$ has only $O(ML^{11/2})$ as the generic doubled cross with (17). No complete fifth-power bound follows from this sector theorem.

9 What the bounds do not establish

The remaining problem is an averaged one-log gain for the actual source, not a pointwise contraction of every carrier fibre. Define $\Theta_U(N)=A_U(N)/(\log N)^2$. A sufficient next target is

$$\Sigma_{U^2<s\le T/4}\,|D_U(s)|^2\,\Sigma_{m\le T/s}K(m)\Theta_U(ms)$$
$$=O(T(1+L_0)^5). \quad (18)$$

The present theorem gives the sixth-power envelope. It does not prove (18).

A proposed uniform $\Theta_U(N)=O(1/\log U)$ fails. Choose eight distinct primes in fixed doubling intervals between t and 256t, set $U=t^{29/10}$, and let N be their product. For large t all pair products are below U/8, all triple products exceed U, and $N\le U^3/8$. Every six-prime cofactor has exactly 20 ordered long 3+3 partitions, each with coefficient one. The admissible cofactor responses therefore agree, and weighted Cauchy is an equality on this family. As $t\to\infty$,

$$\Theta_U(N)=1-\Sigma_i(\log p_i)^2/(\Sigma_i \log p_i)^2 \to 7/8.$$

This refutes the pointwise shortcut, not the averaged target. No substantial density or total energy contribution of this family has been established here.

Likewise, an untruncated harmonic lower bound does not transfer automatically to the finite physical window. The cutoff can destroy the orthogonality of harmonic rows, and their physical images can cancel after recombination. A nonzero projected row alone is not a lower bound for the full energy.

Theorem 2 is cumulative in $N\le T$. It provides no estimate proportional to the length of a thin interval. Neither theorem gives a literal coefficient-one inequality, a sharp asymptotic constant, or a response theorem for an undefined terminal map. In particular, no inference is made to the unbound I37/I54 response construction or to a Walsh model.

For the sector refinement, ownership belongs to complete synthesized outputs. Ordered subtraction of previously paid complete-output classes preserves stronger bounds on those classes. Internal source selectors, Z/J packets, or overlapping external fields cannot be treated as disjoint output projections. The fully squarefree residual remains outside the repeated-decorator argument.

10 Reproducibility and evidentiary status

The accompanying Python program uses only the standard library. Run “python3 verify_energy.py 40” and “python3 verify_energy.py 80” from the unpacked audit folder. No network, private directory, or secret input is required. It constructs μ and literal truncated Möbius coefficients, retains signs, enumerates ordered carrier and long-role histories, and compares their formal polynomials in log p with the compressed K*D formula. It also compares exact signed squares with the gcd Gram parametrization. The common normalizer 3/log N is left outside these polynomial identities.

At U=40, T=8000, the computation has 2 nonzero source outputs and 3 ordered histories. At U=80, T=64000, it has 435 nonzero source outputs, 1298 ordered histories, and 688 ordered Gram terms. Of the latter, 551 have a common gcd d sharing a prime with a,b, or k. These counts describe the specified finite tests, not a statistical sample or asymptotic certificate.

The program rejects an absolute-value replacement of signed source coefficients. Separate synthetic projection fixtures reject a nonstrict lower boundary and a strict upper boundary. They are labeled synthetic because a true source coefficient can vanish at a boundary and therefore fail to expose a faulty comparison. The algebraic 4-versus-8 carrier witness rejects an unjustified additional coprimality condition. Prime-power carrier identities are checked explicitly.

Analytic status is separate. The carrier lemma and harmonic fifth-moment chain are proofs using the fixed external Möbius input. The one- and two-large-prime estimates complete the bare sixth-power result. The first-power repeated-decorator refinement remains conditional on the stated uniform marked-role and physical-map inputs. The executable does not certify those inputs, the imported theorem, asymptotic estimates, or an unspecified downstream response.

References

[1] M. Ram Murty and Akshaa Vatwani. A remark on a conjecture of Chowla. Journal of the Ramanujan Mathematical Society 33(2), 2018, 111–123. Proposition 3, printed page 115. Author manuscript: https://mrammurty-ai.github.io/chowla.pdf. Only the fixed A=4, q=1 consequence is used. The primary statement was independently inspected in this audit; the underlying proof attributed there is imported rather than re-proved.

[2] Finite carrier window complete bare source exponent 6. Research argument dated 23 October 2026, Sections 1–7. Source of the normalization lemma, exact window, and pointwise obstruction.

[3] Joint first order rows close the one log upper gate. Research argument dated 23 October 2026, Sections 1–7. Source of the full harmonic fifth-moment derivation.

References continued

[4] Quantitative signed estimate for the whole no first order sector; Recovered supplied q1 theorem and Abel tail; Required input and exact bounded base two Möbius convolution. Research arguments dated 23 October 2026. These supply the scalar correction, exclusion kernel and fixed-precision application used in Sections 4–5.

[5] Complete one large sector 2; Natural carrier overlap; Actual two large prime source collapse and uniform energy; Ordered log second moment. Research arguments dated 22–23 October 2026. These supply the exact sector decompositions, one-large-prime profile, carrier transfer and prime-pair bound used in Section 6.

[6] First power support one outputs with an owned large decorating repeat. Research argument dated 27 October 2026. The exact owner decomposition, repeat-tail summation, constants and internal cross estimate are reproduced in Sections 7–8, conditional on their identified inputs.

[7] Conditional physical bound with arbitrary marked primes. Predecessor research argument dated 27 October 2026. This is the inherited marked-role and physical-map input for [6]; the present paper does not silently upgrade it to a self-contained proof.

The packet includes portable source transcriptions and a provenance manifest. Historical original-file hashes are reported as source pins, not as hashes of the reformatted transcriptions. The packet separately computes hashes of its own actual files. The dated research notes are unpublished working sources, not peer-reviewed publications. No novelty or independent third-party replication is asserted.

Author

Justin Ayars